

LECTURE SERIES · ISEA

Intelligent Digital Forensics: LLMs, NLP, and Explainable AI for Cyber Investigations

Misinformation, Deepfakes, and Synthetic Media Forensics

Day 2 · Session 4

2 Hours

Hands-On First

Reverse Image · InVID · Fact-Checks

When seeing stops being believing, verification becomes a forensic discipline



SESSION 4 · EXHIBITS



SCAN ME

Exhibit V1 Is Spreading Right Now



Open V1

First item in the exhibits Doc: a dramatic image, captioned as this week's breaking news. Look at it properly — thirty seconds.



Vote, in writing

Real, fake, or can't tell? On campus: hands per option. Online: type V1 = REAL / FAKE / CAN'T TELL in chat. Commit before anyone speaks.



Hold your reason

Note the one thing that drove your vote. By the debrief you will know whether it was evidence — or vibes. Most first votes are vibes.



V1 — as circulating. Working copy: exhibits Doc.

This session's promise: this morning generative AI erased phishing's language tells. Now the same trick attacks eyes and ears — and by the end you will have verified suspicious media with your own hands: reverse search, keyframes, metadata, fact-check workflows.

How the Next 120 Minutes Run



	Block A · Trace the media	Cold vote → 10-min verification toolkit → 25-min five-exhibit lab → reveal	55 min
	Block B · The failed deepfake	10-min Zelensky anatomy → 25-min fact-check workflow + readiness audit → debrief	50 min
	Close-out	Deliverables check, personal-protocol round, takeaways, Day 3 trailer	10 min
	Open floor + buffer	Your questions, spillover — plus transitions and tech recovery	5 min

Bench check: browser only — Google Lens working, InVID-WeVerify installed from the handout link, fact-check sites bookmarked (AltNews, BOOM, Factly, Fact Crescendo, PIB, AFP), exhibits Doc and scoreboard Sheet open in tabs. No code today.

Not All Fakes Are Deep



Cheapfakes / shallowfakes

No AI required

Recontextualised real media (old photo, new caption), selective cropping, speed changes, fabricated screenshots, basic edits. Cheap, fast — and historically the majority of viral visual misinformation.



Manipulated media

Real base, altered

Face swaps and lip-sync reenactment on genuine footage; spliced audio; object removal. A real event, surgically falsified.



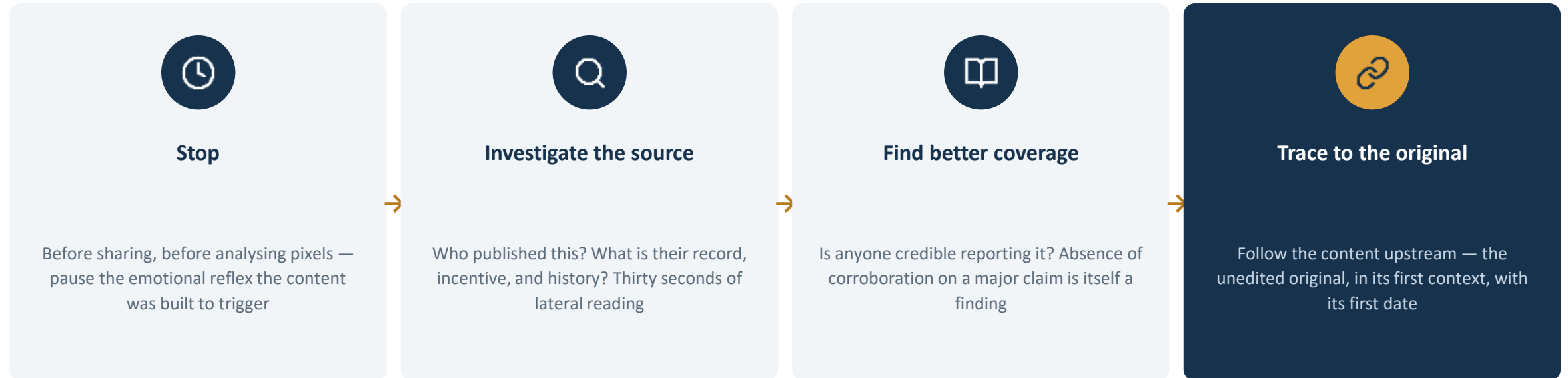
Fully synthetic media

Generated from nothing

Text-to-image and text-to-video, GAN and diffusion faces, cloned voices — no original event existed at all.

Why the taxonomy matters operationally: a cheapfake falls to reverse image search in ninety seconds; a full synthesis never appeared anywhere before. Different fakes, different forensics — misclassify the fake and you pick the wrong tool.

Provenance First: The SIFT Discipline



Why provenance beats pixels: most viral fakes are real media in false context — no pixel analysis detects a true photo with a false caption. Only its history does. Ask where it came from before asking what's wrong with it.

Reverse Image Search: The False-Context Killer

`reverse_image_workflow`

WORKFLOW

1. Right-click image → "Search with Google Lens"
(or upload at `images.google.com`)
2. Review visually similar + exact matches
3. Sort by DATE — find the EARLIEST appearance
4. Open earliest source → original caption,
original date, original location
5. Compare against the claim under test

Also: TinEye (sort by oldest), Yandex, Bing

What it proves — A years-old flood photo attached to this week's 'breaking news' dies the moment its original appearance surfaces — case closed in minutes.

What it can't prove — A fully generated image has no prior history. Zero matches ≠ authentic — it may mean too new, too cropped, or born yesterday in a model.

Craft notes — Search cropped regions separately; try mirrored versions; screenshots of screenshots still match — persistence beats sophistication.

Activity 1 runs exactly this workflow on the exhibit pack — including one image designed to return nothing.

Metadata and the InVID-WeVerify Toolkit



EXIF metadata

Camera model, timestamps, GPS, edit software traces — when present, a provenance goldmine; Session 1's metadata lesson in image form.



The stripping caveat

Major platforms remove EXIF on upload — absence is normal, not suspicious. Original files (from a device, an email attachment) are where metadata survives.



Video keyframes (InVID)

The plugin splits video into representative frames and reverse-searches each — turning the image workflow into a video workflow in one click.



Magnifier & forensic filters

Zoom and enhancement for artefact inspection: boundary checks, text-in-image examination, error-level views — triage aids, never verdicts — the debrief explains why.



Contextual queries

InVID's Twitter/X search, thumbnail lookups, and weather-and-daylight sanity checks — cross-examining the claim's circumstances, not just its pixels.

InVID-WeVerify: free, browser-based, built with fact-checkers — the closest thing verification has to a standard field kit.

Five Exhibits, All Claims Under Test



V1

“Flood devastation, this week” — dramatic image + breaking-news caption



image + claim

V2

“Leaked memo screenshot” — official-looking circular spreading on WhatsApp



screenshot

V3

“Protest crowd” — striking photo, no prior appearances findable



image + claim

V4

“Assam under water, right now” — 40-second flood clip pinned to this week



video

V5

“Shocking but real?” — an image that looks too dramatic to be true



image + claim

Design disclosure: the pack mixes miscaptioned-real, fabricated, AI-generated, AND genuine items — because a verifier who can only say 'fake' is half a verifier.

Reverse Image Verification — V1, V3, V5



Run the workflow

Google Lens / images.google.com on each exhibit; TinEye's oldest-first sort as the cross-check. Cropped-region searches where full-image search stalls.



Hunt earliest appearance

For V1: when and where did this image actually first appear — and what did its original caption say?



Interpret absence carefully

V3 returns nothing. Write down what 'no history' can mean — too new, too cropped, or never real — and which additional check each possibility demands.



Draft calibrated verdicts

One sentence per exhibit in calibrated verdict language: what you can support, on what evidence, with what stated limit — into your scoreboard row.

Prediction on record: V1 falls in under three minutes — and V3 will tempt you to overclaim. Resist in writing.

InVID and Metadata — V2, V4



Keyframe the video (V4)

InVID → Keyframes: fragment the clip, reverse-search the frames. Does this footage predate the claimed event? Different setting? Different language?



Magnify the screenshot (V2)

InVID magnifier on the 'memo': font inconsistencies, alignment drift, seal and letterhead oddities, figures and reference numbers that fail a sanity check.



Read metadata where it lives

The handout's original photo (emailed file): EXIF timestamps, device, GPS — versus its WhatsApp-forwarded twin's silence.



Context checks

V4's checkable circumstances: does the claimed date's weather, venue, and language match what the frames show?

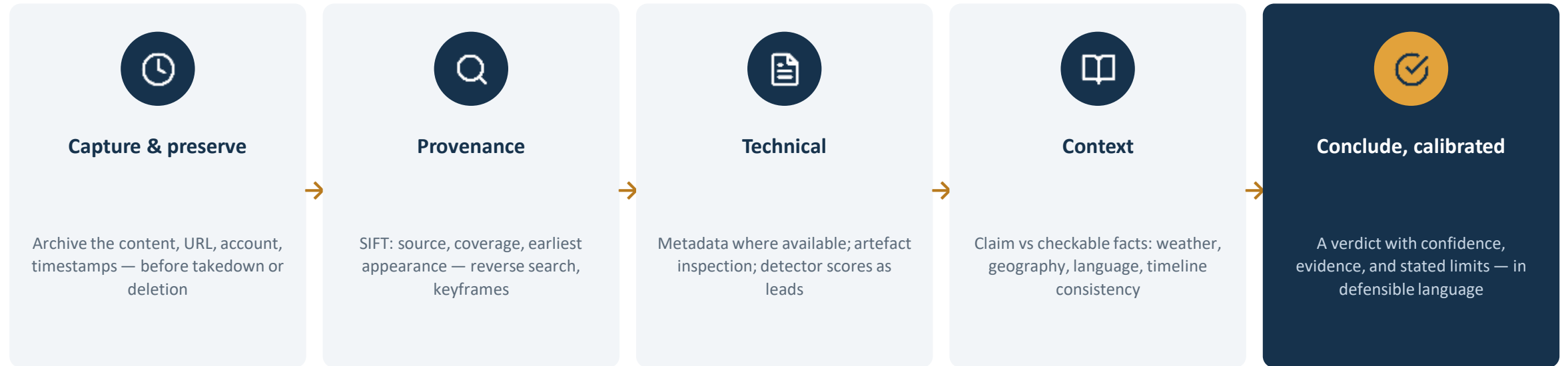
The V4 pattern to expect: the video is real — the caption is the lie. Miscontextualised beats manipulated, again.

SESSION 4 · HANDS ON



SCAN ME

The Verification Workflow — and Its Language



Verdict language, graded in the lab: “Miscaptioned — same image indexed years earlier under a different event (evidence: earliest appearance)” beats “fake”. And the crucial asymmetry: “no evidence of manipulation found” is NOT “authentic” — absence of findings is a bounded statement about your search, never a certificate.

The Reveal — Five Verdicts, Five Lessons



V1 · Miscaptioned real photo

The earliest indexed appearance kills it in minutes: real image, different year, different event. The workhorse fake — and reverse search is its natural predator.



V2 · Fabricated screenshot

The “memo” never existed — fonts, seal, and reference numbers give way under the InVID magnifier. A cheapfake: no AI was required to lie.



V3 · AI-generated image

Zero history plus generation-consistent features supports “consistent with AI generation” — never “proven fake”. If you overclaimed in writing, that discomfort is the lesson.



V4 · Miscontextualised video

The footage is real; the caption lies. Keyframes date it to a different event entirely — pixels honest, context false. Miscontextualised beats manipulated, again.



V5 · Genuine

Shocking and true. If you called it fake on vibes, you just paid the liar's dividend in person. Defending the real, with evidence, is half this profession.

Deepfake Detectors: The Same Hard Truths



How they work

Classifiers trained on known fakes: artefact patterns, frequency fingerprints, physiological signals — genuinely strong on what they trained on.



How they fail

Generalisation collapses on new generators, compression, and re-encoding — a WhatsApp-forwarded video is pre-laundered against detection.



The arms race, again

Every published cue teaches the next generator what to fix — Session 3's text-detector story, frame for frame, in pixels.



The base-rate trap

At internet scale, even 95% accuracy drowns true positives in false alarms — a percentage without a base rate is theatre.



The forensic posture

Detector scores corroborate provenance and context; they never stand alone. 'The detector said fake' is not a finding — it is a lead.

Deliberate symmetry with Session 3's text-detector slide: same epistemics, same posture, different medium. Learn it once, apply it twice.

16 March 2022: The Surrender That Never Happened

-  Amid full-scale war, hackers compromised Ukrainian broadcaster Ukraine 24 — planting a fake surrender message on its website and live ticker — while a deepfake video of President Zelensky telling soldiers to lay down arms circulated on social platforms.
-  Technically poor: a near-static body, mismatched head proportions and skin tone, unnatural voice — artefacts visible to ordinary viewers, not just experts.
-  Weeks earlier, Ukrainian authorities had publicly warned that exactly this — a fabricated surrender video — was coming. The audience had been inoculated in advance.
-  Zelensky rebutted within hours via his own established channels — the same daily-address format his public knew intimately. Platforms removed the video under manipulated-media policies.
-  The attack failed almost completely: minimal belief, rapid global debunking — and instant status as the textbook case you are studying right now.

Why It Failed — Read Carefully



Prebunking worked

The audience had been told this fake was coming. Inoculation — not detection technology — did the heaviest lifting. Nobody ran a classifier; everybody remembered the warning.



Channel authentication won

The rebuttal's power was WHERE it came from: the president's own verified, habitual channels. Identity was proven by provenance, not by pixels — the session's whole argument, in one move.



Speed sealed it

Hours, not days. The correction ran on the rumour's own timescale, through the rumour's own platforms — the correction-science checklist, executed under fire.

The uncomfortable counterfactual: run the same attack with 2026-quality synthesis, no prior warning, and a target without trusted rapid channels — a company, a candidate, a community leader. Which of the three defences survives? Only the ones you built in advance. That is the case study's real assignment.

Seeing Is No Longer Believing — Both Ways



The obvious harm

fake accepted as real

- Fabricated events enter the record: false statements, false crimes, false scandals
- Reputations, markets, and elections moved by things that never happened
- The harm everyone anticipates — and defends against



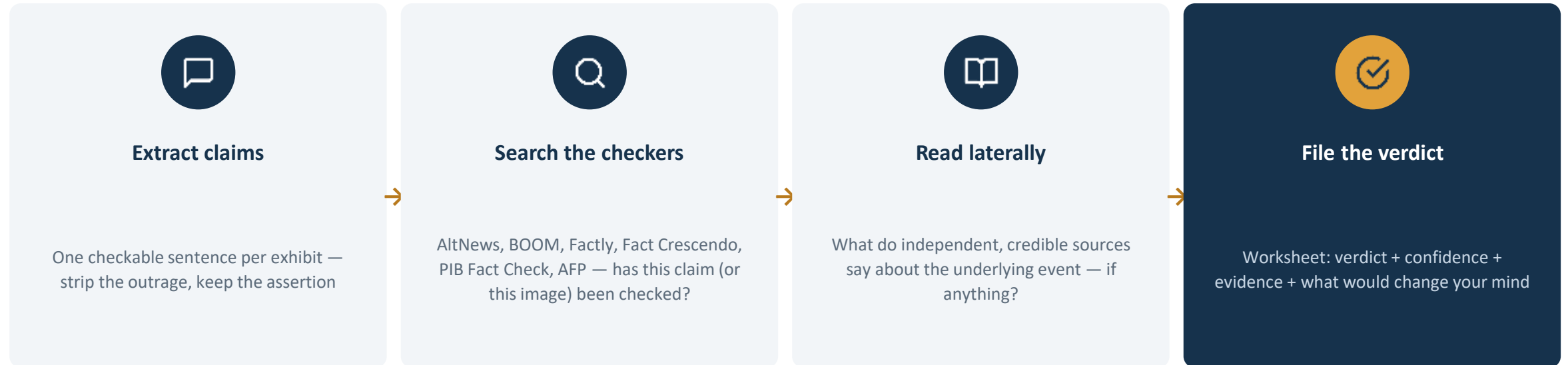
The liar's dividend

real dismissed as fake

- Genuine evidence denied: “that recording is a deepfake” becomes a universal defence
- As public awareness of deepfakes grows, the dividend grows with it
- Scholars Chesney and Citron named it — arguably the deeper long-term injury to justice

The forensic mandate this creates: we now must be able to prove authenticity as rigorously as we prove fakery. Verification is no longer only about catching fakes — it is about defending the real.

The Full Fact-Checking Workflow — All Five



The V5 test — defend the real: one exhibit is genuine. Verifying it as genuine — with evidence, not vibes — is the liar's-dividend skill this whole session has been building toward. 'Shocking' and 'false' are different properties; your worksheet should prove you know that.

Bring It Home



Question 1 · Transpose

A fabricated audio clip of a local official announcing a curfew hits your district's WhatsApp networks tonight. Walk the first six hours: who detects, who verifies, who rebuts — through what channel?



Question 2 · Readiness

Score your organisation on the Zelensky trio: prebunked audience? authenticated rapid channel? response playbook? Most score one of three — which is your missing pair?



Question 3 · The dividend

A genuine, incriminating recording surfaces in one of your cases — and the subject declares it a deepfake. What does your side now have to prove, and could you?

SESSION 4 · HANDS ON



SCAN ME

Prebunking: Inoculation Before Exposure

-  **The inoculation idea**

Expose people to a weakened dose — the manipulation technique, named and demonstrated — before the real thing arrives. Psychological vaccination, literally the research metaphor.
-  **Technique beats topic**

Teaching 'this is how false-context images work' generalises across every future rumour; debunking one rumour dies with that rumour.
-  **The evidence base**

Inoculation studies (Cambridge and others) and large platform prebunking campaigns show measurable, durable resistance — modest, real, scalable.
-  **The Zelensky proof**

Ukraine pre-warned its public that surrender deepfakes were coming — weeks early. When the fake arrived, the audience had antibodies. The case study, revisited.
-  **Your organisational version**

Brief staff on the current quarter's fraud patterns before the attacks land — threat intel repurposed as inoculant. Cheap, and it compounds.

Detection reacts; verification investigates; prebunking prevents. A mature posture runs all three.

Investigators as Trust Infrastructure



Your verdicts carry weight

A forensic professional saying 'authentic' or 'manipulated' moves institutions — which is precisely why calibrated language is an ethical duty, not style.



Verification power cuts both ways

The same skills that debunk can lend false credibility — 'experts confirmed' is itself a manipulation trope. Guard your name like evidence.



Show your working

Publishable methodology — the SIFT trail, the earliest-appearance evidence — lets others check you, which is where public trust actually comes from.



Honest limits, stated

'We could not determine' is a professional finding. The field's credibility dies faster from overclaiming than from admitted uncertainty.

The quiet thesis of the whole series: in a synthetic-media world, disciplined verifiers ARE the trust layer. Train accordingly.

Deliverables and Debrief



Deliverable · Verification report

Five calibrated verdicts with evidence trails and stated limits — your scoreboard row, completed in the workflow's language.



Debrief 1 · Propagation

For the fakes you caught: where would each spread fastest in your context, who would believe it, and who is positioned to correct it — through which channel?



Debrief 2 · Trust

The closing round: after today, what is YOUR personal protocol when shocking media reaches you — and what should your organisation's be?

The syllabus outcome, checked: emerging AI-generated threats understood — and detection practised with your own hands. Both boxes, ticked by you.

Key Takeaways

1

Most fakes are cheap: recontextualised reality outnumbers deep synthesis — and provenance-first verification catches both.

2

The liar's dividend cuts both ways: 'it's a deepfake' now shields the guilty — defending the real is as much the job as catching the fake.

3

Detectors are leads, provenance is proof: SIFT, reverse search, and metadata beat pixel-gazing — and 'no evidence found' is not 'authentic'.

4

Trust is becoming infrastructure: prebunked audiences, authenticated channels, rapid response — the Zelensky trio is the working playbook.



UP NEXT · SESSION 5

Explainable AI (XAI) for Intelligent Cyber Investigations

Day 3: the explanation machinery itself — SHAP, LIME, and attention, computed by you, on the classifiers this series keeps promising to open up.

Questions & discussion

“Assam Is Drowning — This Week”

As received: *“Assam is drowning. A father carries his children to safety through neck-deep water — this week. Share to spread awareness.”*



Task: Reverse image search. When and where did this image actually first appear — and what did its earliest caption say?

“Leaked Before the Announcement”

As received: “Petrol up ₹10/litre, diesel ₹12.50/litre — order + Finance Ministry notification attached.”

MINISTRY OF PETROLEUM & NATURAL GAS
(पेट्रोलियम एवं प्राकृतिक गैस मंत्रालय)

ORDER
New Delhi, the 29th April, 2026

Due to the sharp increase in global crude oil prices by over 100% in recent months, the cost of procurement has risen significantly, directly impacting domestic pricing. As a result, Public Sector Oil Marketing Companies (OMC), including Indian Oil Corporation, Bharat Petroleum and Hindustan Petroleum, are incurring substantial under-recoveries at the retail selling price level, quantified at approximately Rs. 24.40 per litre on petrol and Rs. 104.99 per litre on diesel as on 01.04.2026.

Such sustained under-recoveries are economically unsustainable and adversely impact the financial viability and operational capacity of the OMCs, which are responsible for ensuring uninterrupted supply and distribution of petroleum products across the country. Continued suppression of retail prices in the face of escalating input costs distort market-linked pricing mechanisms and result in accumulation of losses, thereby necessitating corrective pricing measures.

In view of the above, a calibrated upward revision of the retail selling price of petrol and diesel is immediately warranted to partially offset the prevailing under-recoveries, restore financial stability of the OMCs, and ensure continuity of supply.

Sl. No.	Description of Item	Upward Revision
1.	Motor Spirit, commonly known as Petrol	Rs. 10/- per litre
2.	High Speed Diesel	Rs. 12.50/- per litre

This order shall come into force with immediate effect.


Bhaskaran Ganesan, DD(OMC)

MINISTRY OF FINANCE
(Department of Revenue)
NOTIFICATION

New Delhi, the 29th April, 2026
No. 13/2026 Central Excise

G.S.R. 275(0)- WHEREAS, the Central Government on being satisfied that the Special Additional Excise Duty leviable under Section 147 of the Finance Act, 2002 (20 of 2002) on:

- (i) Motor Spirit commonly known as petrol; and
- (ii) High Speed Diesel oil,

should be increased and that circumstances exist which render it necessary to take immediate action.

NOW, therefore, in exercise of powers conferred by Section 7B of Central Excise Act, 1944 (1) of 1944 read with Section 147 of the Finance Act, 2002, the Central Government, hereby directs that the Eighth Schedule to the said Finance Act, 2002, shall be amended in the following manner, namely: -

In the Eighth Schedule to the Finance Act, 2002, (i) against Item No. 1, for the entry in column (3), the entry "Rs. 11 per litre" shall be substituted; and (ii) against Item No. 2, for the entry in column (3), the entry "Rs. 12.50 per litre" shall be substituted.

This notification shall come into force with immediate effect.

[F. No. 191408-13/2026-TRU]

DIREKHAJ SHARMA
Under Secretary to the Government of India

Task: InVID magnifier on both documents: fonts, seals, reference numbers, figures that fail a sanity check. Then find better coverage.

“Mainstream Media Won’t Show You This”

As received: “Massive crowds at yesterday’s protest — mainstream media won’t show you this.”



Task: Reverse search it. Write down carefully what your result does and does not prove.

“Assam Right Now — 4 August”

As received: “Assam right now (4 August) — entire towns under water.”



archive.org/details/20260805-0729-32.9925298

Task: Open the link → InVID keyframes → reverse search the frames. Do the claimed date, place and language match what the frames show?

“Wading Out With Whatever He Could Carry”

As received: *“Wading out with whatever he could carry — Assam floods, this week.”*



Task: Same discipline as the rest. Prove genuine or prove false — with evidence, not vibes.